

Final Project

Engr 1201

Brendan Spires

Objective:

To construct a Savonius Style Vertical Axis Wind Turbine (VWAT) with a structural footprint of less than 3ft cubed capable of generating a stable current in low wind to turbulent wind conditions. The energy of which would be stored in a portable battery bank and used to charge a phone. All of this while using minimal tools and keeping costs low.

Design:

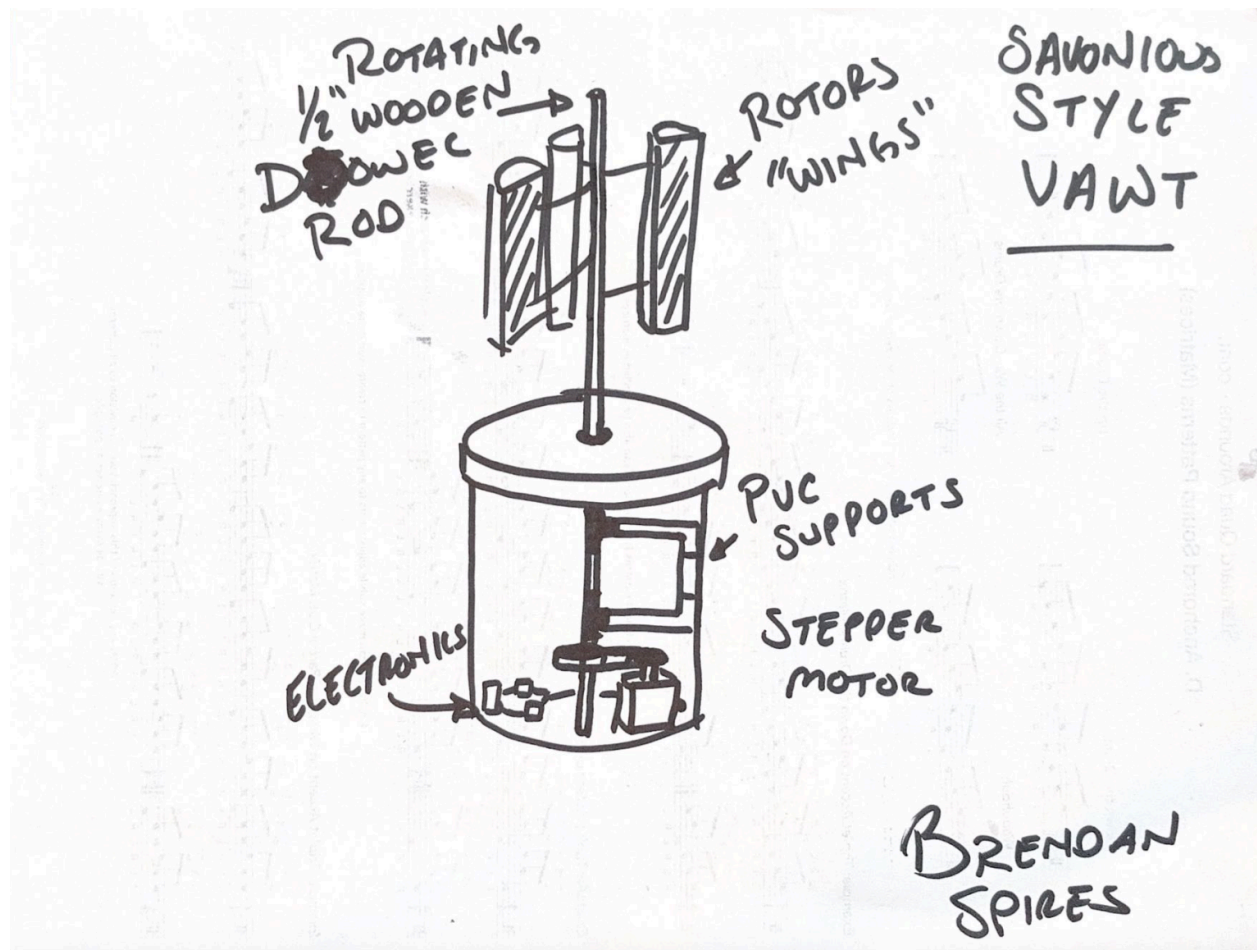


Figure 1 : Basic diagram of the VAWT design

The design is simple, a plastic enclosure to shield the electrical components from the weather with a PVC support system holding up a wooden dowel rod at the center. Attached to this dowel rod is a ½” timing pulley with a timing belt that runs to a second timing pulley attached to the top of a stepper motor. Those wires run through a series of electrical components which clean up the choppy electrical current and deposit that electricity into a battery bank. That power can then be used to charge a phone and what not. To spin the dowel rod, half cylinder rotors will be attached to catch incoming wind from any direction. This design is meant for low wind and low starting torque.

Results:

Failures:

- As I could not find materials to use for the rotors I ended up cutting two large plastic bottles in half and fashioning them to the center beam. Their flexibility eliminates any ability to capture high winds
- When the lid is on, it compresses the PVC which in turn compresses the wooden dowel rod. This makes it much harder to turn so the starting torque would have to be significant. This defeats the “low torque startup” I was expecting.
- The stepper motor doesn’t turn as easily as I expected. This in combination with the previous two means the central rod does not turn so no electricity is produced. This is the biggest failure of the project.
- The timing belt must be under a constant high tension or it won’t grab the teeth of the pulleys. This has resulted in slippage if the central rod is turned too abruptly.

Successes:

- I have a much greater understanding of electronics and current. There are several processes of taking a raw unstable current and smoothing it out to something more usable.
- I also got great soldering practice. It has been close to a decade since ive used one and it was such a rewarding experience to solder two wires together successfully.
- I also got experience with a multimeter, every time I would solder I would check continuity with the multimeter to make sure the connection was good and I could move on.
- When turned fast enough the motor actually generates a current of around 4V which lights up the buck converter display.

I have spent the better part of a month gathering the materials, putting together a plan and executing it. Although the project as a whole is a failure on paper, being that it did not fulfill its intended purpose, I wouldn't consider the whole experience a wash. All of my successes were in learning something new. This has ignited a desire to learn more and experiment more with electronics in the future.

In the future should I attempt this project again, here is what I'd do differently.

- Instead of using a belt and pulleys I would have a system of interlocking gears. This nearly eliminates the slippage and tension problem. It also generates more mechanical advantage which eases the torque requirements to turn the turbine.
- For the scoops, having a fully 3d printed set which lock into positions 120° apart with that same 1/2" bore would instantly solve the wind catching issue.
- Instead of using a cylindrical housing, finding a rectangular housing for everything would increase stability and make attaching components to the walls much easier.

For further information, I kindly request you visit this github repo where i have documented my journey.

Again, this has been a fun project and I have learned so much. This has been an altering experience, I am now more excited for electronics projects and have overcome the perfection blockade. Now I just want to create. I'm already planning another in my near future and cannot wait to document that one as well.